Joint Low-Rank and Smooth Tensor Regularization for Multispectral and Hyperspectral Image Fusion

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***Abstract*—The fusion of multispectral and hyperspectral images (HSIs) has become an essential technique for generating images with simultaneously high spatial and spectral resolutions. However, this task is inherently an ill-posed inverse problem, and its solution typically relies on incorporating prior knowledge to constrain the search space. Conventional approaches often adopt an additive combination of low-rankness and smoothness priors [e.g., minimization of nuclear norm and total variation (TV)]. While effective to a certain extent, this additive formulation fails to fully capture the intrinsic coupling between the two priors and is highly sensitive to the choice of tradeoff parameters. To address these issues, we propose a weighted low-rank and smooth tensor (WLRST) regularization framework for fusing multispectral and HSIs. Our WLRST leverages the fact that the original image and its gradient maps share similar manifold structures while exhibiting different smoothness levels across spatial–spectral modes. It imposes tensor nuclear norm (TNN) constraints on multimode gradient maps, with adaptively assigned weights that reflect the smoothness variation in each mode. This unified regularizer jointly embeds low-rank and smoothness properties, allowing more precise exploitation of spatial and spectral correlations in the fused image without extensive parameter tuning. Experimental results on benchmark datasets demonstrate that our WLRST method produces significantly finer fusion results compared with state-of-the-art techniques, achieving superior visual quality and improvements across multiple quantitative metrics.**

***Index Terms*—Multispectral and hyperspectral image (HIS) fusion, tensor nuclear norm (TNN), total variation (TV), weighted low-rank and smooth tensor (WLRST) regularization.**

I. Introduction

HYPERSPECTRAL imaging offers the unique capability to capture rich, contiguous spectral information across a wide range of wavelengths. Hyperspectral images (HSIs) enable precise material identification and quantitative analysis, making them invaluable in various applications such as precision agriculture, environmental monitoring, geological surveying, and urban planning [1], [2], [3], [4]. However, there exists a fundamental tradeoff between spectral and spatial resolutions in optical imaging systems [5]. HSIs provide high spectral resolution at the expense of low spatial resolution (LR), which is complementary to multispectral images (MSIs) that contain fewer and broader spectral bands but typically offer much higher spatial resolution (HR) [6], [7]. In recent years, the fusion of an HR-MSI and an LR-HSI of the same scene has emerged as an effective strategy to overcome these limitations and obtain an HR-HSI that preserves the discriminative spectral information from the HSI while enhancing spatial details from the MSI [8].

MSI–HSI fusion is a severely ill-posed problem, as MSIs and HSIs are subject to different degradation processes, resulting in complementary yet irreversible spatial–spectral information loss [9]. Consequently, recovering an HR-HSI from such heterogeneous observations is inherently nonunique and highly sensitive to model mismatch. This difficulty is further compounded by the distinct correlations and anisotropic characteristics across spatial and spectral dimensions, making effective prior modeling essential for accurate reconstruction. Existing techniques for HSI-MSI fusion can be broadly categorized into model- and learning-based approaches [10], [11]. Convolutional neural network (CNN)-based methods typically adopt 2-D or 3-D convolutional architectures to jointly exploit spatial and spectral information for HSI–MSI fusion. SSMSFuse proposes a CNN-based two-branch fusion model that employs a multiscale CNN (Spa-Net) to extract spatial features from MSI and a self-attention spectral branch (Spe-Net) to capture HSI spectral dependencies [12]. SSRFN employs a convolutional backbone augmented with Mamba modules to jointly exploit cross-modal local and global features [13]. More recently, attention mechanisms and transformer-based architectures have been introduced to capture long-range spatial–spectral dependencies. SEST is a spectral-enhanced sparse transformer network for HSI superresolution that combines a sparse transformer to efficiently capture nonlocal spatial similarities with a spectral enhancement module to exploit low-rank spectral characteristics, both integrated into a multiwindow residual block to adaptively fuse fine-grained spatial–spectral features [14].

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MIMFormer is a multiscale Inception mixer transformer that integrates CNNs and transformers to capture local and global spatial–spectral features via spectral splitting, convolution, pooling, and self-attention, thereby enriching details in fused images [15]. Furthermore, the modality-aware feature-aligned network [16] addresses modality mismatch in pan-sharpening, CasFormer [17] employs cascaded transformers to capture spectral continuity and enhance spatial detail in computational hyperspectral imaging, and MambaX [18] proposes a nonlinear state predictive control model for image superresolution. These studies demonstrate the strong potential of deep learning-based methods in modeling complex spatial–spectral relationships. At the same time, such deep models usually require large-scale training data, and their generalization ability may depend on the availability of representative training data across sensors and acquisition conditions [19], [20], [21]. In contrast, model-based methods typically formulate a physically motivated observation model that links the unknown HR-HSI to the observed LR-HSI and HR-MSI through spatial blurring, downsampling, and spectral response operators and then regularize the solution using appropriate image priors. Owing to their explicit priors and physics-based observation models, model-based approaches remain highly attractive for many remote sensing applications.

In model-based approaches, matrix factorization-based methods have been widely used for MSI and HSI fusion, typically reshaping the HR-HSI into a matrix that is factorized into a basis (endmember) matrix estimated from the LR-HSI and a coefficient (abundance) matrix inferred from the HR-MSI. CNMF is a coupled nonnegative matrix factorization framework that imposes shared endmember and abundance matrices for HSI and MSI while embedding sensor observation models into the factorization [22]. CSU performs hyperspectral superresolution by jointly unmixing the input images into endmember spectra and abundance maps through a coupled matrix factorization framework constrained by basic physical properties of spectral mixing [23]. HySure formulates fusion as a convex optimization problem and incorporates vector total variation (TV) to capture joint spatial–spectral smoothness, thereby preserving edges while handling blur, resolution differences, and noise [24]. A variational fusion method formulates the task as an inverse problem in a low-dimensional subspace with sparsity enforced via dictionary atoms and the corresponding coding coefficients learned from the observed images [25]. Subsequently, a fast multiband fusion algorithm based on solving a Sylvester equation has been developed on this basis [26]. More recently, the WNNSTV framework imposes weighted nuclear norm minimization, ℓ_1-norm, and TV regularization on the abundance matrix to enhance spatial–spectral sparsity and promote piecewise smoothness [27]. Although matrix factorization-based methods have achieved remarkable performance, unfolding 3-D HSIs into 2-D matrices inevitably destroys their intrinsic spatial–spectral tensor structure and may limit the ability to exploit high-order correlations.

To better preserve the multidimensional structure of HSIs, tensor representations that more accurately capture and encode the inherent structural information of multidimensional data than conventional matrix representations have been extensively explored for fusion and superresolution. CSTF reformulates the fusion problem as a coupled tensor factorization to jointly estimate three mode dictionaries and a sparsity-regularized core tensor [28]. SSGLRTD formulates the fusion problem as a low-rank tensor decomposition with spatial and spectral graph regularization, jointly exploiting hyperspectral low-rankness and preserving spatial correlations and spectral structures in the fused image [29]. LTTR clusters similar HR-MSI cubes, transfers this structure to HR-HSI to form nonlocally grouped 4-D tensors, and then imposes a low tensor-train rank prior to jointly exploit spatial, spectral, and nonlocal correlations within a TT-rank–regularized superresolution framework [30]. LNTM exploits the shared local and nonlocal manifold structures of HR-MSI and HR-HSI patches to build orthogonal bases with an adaptive truncation that preserves low-frequency content and suppresses noise, and integrates this manifold–truncation regularizer into a variational superresolution framework [31]. HTSR maps tensor factors into a high-dimensional Hankel subspace and imposes low-rank and ℓ_1-norm sparsity regularization, solved via vector–tensor operators and alternating optimization, to exploit prior features, suppress noise [32]. GTNN extends the traditional third-mode tensor nuclear norm (TNN) to arbitrary modes via Fourier transform and, combined with low-rank spectral basis representation and nonlocal patch grouping, more fully exploits multimode correlations and spectral–spatial self-similarities in the fusion process [33]. However, these tensor-based fusion methods often rely on a single prior or decouple spatial and spectral structures and enforce them via multiple regularization terms, which complicates optimization and hyperparameter selection and may limit reconstruction quality.

The fusion problem of MSIs and HSIs is commonly addressed by formulating an optimization model with appropriate image prior regularization [9]. Low-rankness and local smoothness are complementary priors that model global spatial–spectral redundancy and local continuity/piecewise homogeneity, and are usually enforced by nuclear-norm and TV regularizations combined additively [34]. In practice, low-rankness and smoothness are inherently coupled and highly sensitive to the chosen tradeoff between them. Moreover, the original image and its gradient maps share similar manifold structures, leading to highly consistent low-rank properties [35], [36]. At the same time, gradients of HSIs along the two spatial dimensions and the spectral dimension usually exhibit distinct levels of smoothness continuity. This suggests that it is more appropriate to model low-rankness and smoothness in the gradient-tensor domain and adaptively regularize different tensor modes, rather than enforcing the same regularization strength along the spatial and spectral directions.

Motivated by the advantages of tensor representation and correlated TV regularization in hyperspectral image processing, as well as the above observations, we propose a novel MSI–HSI fusion framework based on a joint low-rank and smooth tensor prior. Specifically, we design a weighted low-rank and smooth tensor (WLRST) regularization that imposes a TNN on the multimode gradients of the target HR-HSI while assigning distinct weights to each mode. By constructing a multimode gradient tensor that jointly encodes spatial and

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spectral derivatives, the proposed WLRST prior integrates low-rankness and local smoothness into a single unified regularizer in the gradient domain. This formulation simultaneously exploits the manifold similarity between the image and its gradients, and the heterogeneous smoothness characteristics across spatial and spectral modes. Unlike many existing methods that combine nuclear norm and TV regularization in an additive manner, the proposed method directly couples low-rankness and smoothness in the 3-D gradient-tensor domain. In particular, low-rank constraints are imposed on gradients along all three modes, while modewise weights are introduced to adaptively account for spatial–spectral anisotropy, thereby providing more adaptive spatial–spectral regularization and stronger prior constraints for preserving both spatial details and spectral fidelity in this ill-posed reconstruction task.

The remainder of this article is structured as follows. Section II introduces the proposed fusion framework and describes the WLRST regularization in detail. Section III presents comprehensive experiments on two hyperspectral datasets and analyzes the performance of the proposed approach against state-of-the-art methods. Finally, Section IV summarizes the main findings and discusses possible directions for future work.

II. Proposed Method

In this section, we present the proposed MSI–HSI fusion method based on the weighted low-rank and smooth tensor (WLRST) prior.

A. Formulation of the Fusion Problem

The goal of HSI–MSI fusion is to reconstruct an image that simultaneously possesses high spatial and high spectral resolution by exploiting the complementary information of an LR-HSI and an HR-MSI acquired from the same scene. Let $\mathcal{X} \in \mathbb{R}^{W\times H\times S}$ be the target HR-HSI represented as a 3-D tensor, where W and H denote the width and height of the spatial domain, respectively, and S is the number of spectral bands. The observed LR-HSI and HR-MSI are denoted by $\mathcal{Y}_h \in \mathbb{R}^{w\times h\times S}$ and $\mathcal{Y}_m \in \mathbb{R}^{W\times H\times s}$, respectively. Equivalently, $\mathcal{Y}_h$ and $\mathcal{Y}_m$ can be regarded as spatially and spectrally degraded versions of $\mathcal{X}$, and their relationship can be written as follows:

$$\mathcal{Y}_h = \mathcal{X} \times_1 \mathbf{D}_1 \times_2 \mathbf{D}_2 \tag{1}$$

$$\mathcal{Y}_m = \mathcal{X} \times_3 \mathbf{R} \tag{2}$$

where $\mathbf{D}_1 \in \mathbb{R}^{w\times W}$, $\mathbf{D}_2 \in \mathbb{R}^{h\times H}$, and $\mathbf{R} \in \mathbb{R}^{s\times S}$ denote the downsampling operators along the width, height, and spectral dimensions, respectively. In particular, $\mathbf{D}_1$ and $\mathbf{D}_2$ characterize the spatial response of the imaging sensors, while $\mathbf{R}$ is the spectral response matrix.

Equation (1) means that the point spread function (PSF) of the hyperspectral imaging sensor and the downsampling matrices of the width mode and height modes are separable. Let $\mathbf{M} \in \mathbb{R}^{WH\times wh}$ be the degradation operator of blurring and downsampling with respect to the LR-HSI, and then, the action of the spatial subsampling matrix $\mathbf{M}$ can be decoupled into the two spatial modes of $\mathcal{X}$

$$\mathbf{M} = (\mathbf{D_2} \otimes \mathbf{D_1})^T \tag{3}$$

where $\otimes$ denotes the Kronecker product between matrices. By unfolding these tensors along the third mode, (1) and (2) can be matrixed as follows:

$$(\mathbf{Y}_h)_{(3)} = \mathbf{X}_{(3)}\mathbf{M} \tag{4}$$

$$(\mathbf{Y}_m)_{(3)} = \mathbf{R}\mathbf{X}_{(3)} \tag{5}$$

where $(\mathbf{Y}_h)_{(3)} \in \mathbb{R}^{S\times wh}$, $(\mathbf{Y}_m)_{(3)} \in \mathbb{R}^{s\times WH}$, and $\mathbf{X}_{(3)} \in \mathbb{R}^{S\times WH}$ are the mode-3 unfolding matrices corresponding to $\mathcal{Y}_h$, $\mathcal{Y}_m$, and $\mathcal{X}$, respectively. Under this observation model, the target HR-HSI $\mathcal{X}$ can be recovered by fusing $\mathcal{Y}_h$ and $\mathcal{Y}_m$, leading to the following fusion formulation:

$$\min_{\mathbf{X}_{(3)}} \left\|(\mathbf{Y}_h)_{(3)} - \mathbf{X}_{(3)}\mathbf{M}\right\|_F^2 + \left\|(\mathbf{Y}_m)_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 \tag{6}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm. Because the optimization problem (6) is ill-posed, directly solving (6) usually yields unsatisfactory reconstructions, as the model does not exploit any prior knowledge about $\mathcal{X}$. Therefore, it is necessary to introduce regularization based on reliable prior information to reduce the solution space. In what follows, we analyze the intrinsic structures of the HR-HSI and derive appropriate regularization terms to be incorporated into the formulation (6).

B. Weight Low-Rank and Smooth Tensor Regularization

TV is a well-established regularization technique aimed at preserving spatial smoothness in images. For hyperspectral imagery, the anisotropic form of TV applies the $\mathcal{L}_1$ norm, whereas the isotropic version employs the Frobenius norm to penalize the spatial gradients of the HSI. The TNN serves as another common prior, encouraging a low-rank structure and, thereby, capturing the overall spatial redundancy of the hyperspectral cube. For a structured tensor that simultaneously exhibits low-rank and smooth properties, previous studies commonly adopt an additive formulation, combining two separate penalty terms to represent these priors [35], [36]. As shown in Fig. 1, the gradients of HSIs along all three modes exhibit notable low-rank properties. The singular value distributions further reveal that the rank varies across different gradient directions, indicating anisotropic structural redundancy within the data. Therefore, we design a unified regularizer that imposes low-rank priors on the gradient tensors and adaptively assigns individual weights to each mode, aiming to effectively capture the structural similarity and continuity of hyperspectral data within a single formulation. The weight low-rank and smooth tensor regularization for MSI and HSI fusion can be formulated as follows:

$$\|\mathcal{X}\|_{\text{WLRST}} = \sum_{t=1}^{3} \alpha_t \|\nabla_t(\mathcal{X})\|_{\circledast} \tag{7}$$

where α_t denotes the weight that adjusts the contribution of constraints along different modes, $\nabla_t(\mathcal{X})$ denotes the gradient of $\mathcal{X}$ along the tth mode, and $\|\cdot\|_{\circledast}$ denotes the TNN, defined as follows [37]:

$$\|\mathcal{T}\|_{\circledast} = \sum_{i=1}^{\min(N_1,N_2)} \sum_{j=1}^{N_3} \hat{\mathcal{S}}(i,i,j) \tag{8}$$

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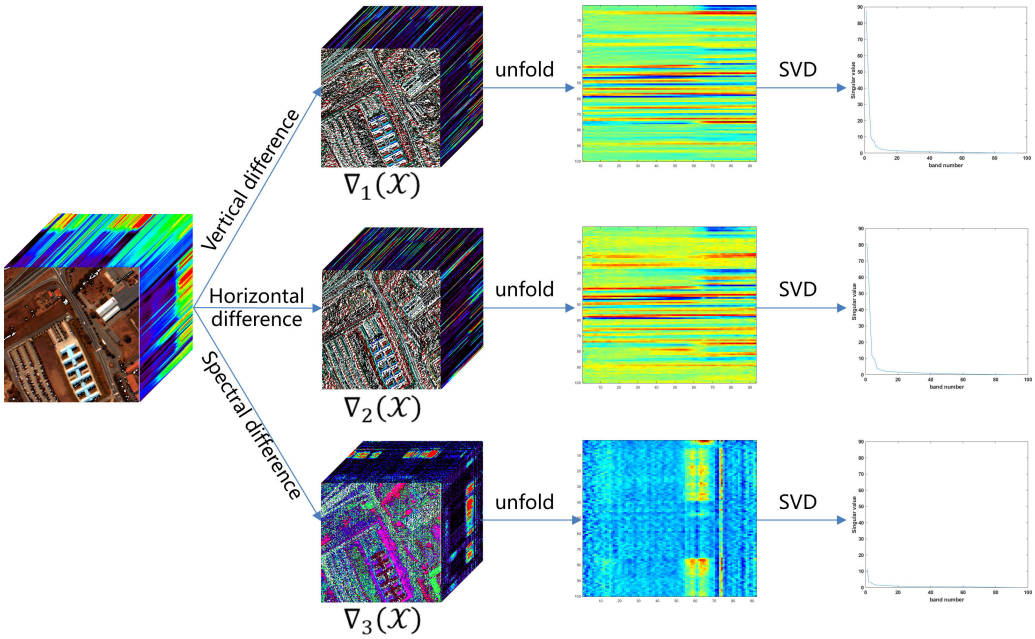


Fig. 1. Rank of gradients along three modes of HSIs.

where $\mathcal{T} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ can be any tensor. Define $\mathcal{U} \in \mathbb{R}^{N_1 \times N_1 \times N_3}$ and $\mathcal{V} \in \mathbb{R}^{N_2 \times N_2 \times N_3}$ are orthogonal, $\mathcal{S} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ is consist of diagonal frontal faces, and $(:, i, :)$, $(:, :, i)$, and $(i, i, :)$ correspond to the ith lateral face, the ith frontal face, and the ith mode-3 fiber, respectively. Based on the above definition, the tensor singular value decomposition (t-SVD) of tensor $\mathcal{T}$ can be defined as follows [37], [38]:

$$\mathcal{T} = \sum_{i=1}^{\min(N_1, N_2)} \mathcal{U}(:, i, :) * \mathcal{S}(i, i, :) * \mathcal{V}(:, i, :)^{\mathrm{T}}. \tag{9}$$

Let $\hat{\mathcal{T}}$ denote the tensor $\mathcal{T}$ in the Fourier domain, and let $\hat{\mathbf{T}}_{(i)}$ denote the ith frontal face of $\mathcal{T}$ in the Fourier domain. The SVD of $\hat{\mathbf{T}}_{(i)}$ is written as $\hat{\mathbf{T}}_{(i)} = \hat{\mathbf{U}}_{(i)}\hat{\mathbf{S}}_{(i)}\hat{\mathbf{V}}_{(i)}^{\mathrm{T}}$. In turn, $\hat{\mathcal{S}}$ in (8) can be obtained by performing an inverse Fourier transform on $\hat{\mathbf{S}}$.

To adaptively balance the contribution of the low-rank constraints along different modes, the weight α_t in (7) is determined by the relative structural redundancy within the gradient tensor $\nabla_t(\mathcal{X})$. The gradient domain represents edges, local transitions, and structural variations more explicitly than the original image, making it better suited for preserving spatial details in fusion. In contrast, direct low-rank modeling on the image tensor mainly captures global redundancy and may be less effective for handling localized structural information. For mode t, the Frobenius norm of its gradient tensor is $G_t = \|\nabla_t \mathbf{X}\|_F$, where $\nabla_t \mathbf{X}$ is the matrix form of $\nabla_t(\mathcal{X})$. Modes with smaller G_t usually exhibit smoother structures and, thus, stronger low-rankness. The weights α_t are computed as follows:

$$\alpha_t = \frac{(G_t + \varepsilon)^{-1}}{\sum_{q=1}^{3} (G_q + \varepsilon)^{-1}}, \quad t = 1, 2, 3 \tag{10}$$

where $\varepsilon > 0$ is a small stabilizer to avoid division by zero. The weight α_t is initialized from the gradient energy statistics of the observed data and adaptively updated at each iteration according to the current estimate of the fused image. This dynamic update allows the weighting scheme to better reflect the evolving spatial–spectral structure during optimization without manual tuning. The adaptive weighting mainly helps balance the contributions of different tensor modes according to their structural characteristics, and the weights are normalized to a comparable scale across modes.

The weighted design provides an additional advantage by adapting the contribution of different tensor modes according to their structural characteristics, rather than treating all modes equally. Compared with existing TNN-based methods, which mainly enforce global low-rankness on the image tensor or its unfoldings, the proposed formulation is more closely linked to structural information relevant to fusion quality. This helps achieve a better balance between spatial detail preservation and spectral fidelity.

The WLRST regularization is introduced into the problem (6), and the novel fusion model is formulated as follows:

$$\min_{\mathbf{X}_{(3)}} \left\|(\mathbf{Y}_h)_{(3)} - \mathbf{X}_{(3)}\mathbf{M}\right\|_F^2 + \left\|(\mathbf{Y}_m)_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 + \lambda \sum_{t=1}^{3} \alpha_t \|\nabla_t(\mathcal{X})\|_{\circledast} \tag{11}$$

where λ is a nonnegative parameter regularization. The proposed fusion model flexibly encodes the low-rank and smooth prior from spatial and spectral dimensions to exploit the potential structure information.

C. Optimization

The proposed model (11) is a variational optimization problem that can be solved in the framework of alternating direction method of multipliers (ADMM) [39]. By introducing these variables $\mathcal{U}_1 = \mathcal{X}$, $\mathcal{U}_2 = \mathcal{X}$, $\mathcal{U}_3 = \mathcal{X}$, $\mathcal{V}_1 = \nabla_1(\mathcal{U}_1)$,

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$\mathcal{V}_2 = \nabla_2(\mathcal{U}_2)$, and $\mathcal{V}_3 = \nabla_3(\mathcal{U}_3)$, the augmented Lagrangian function related to the model (11) can be expressed as follows:

$$\begin{aligned}&\mathcal{L}(\mathcal{X},\mathcal{U}_1,\mathcal{U}_2,\mathcal{U}_3,\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\zeta_1,\zeta_2,\zeta_3,\tau_1,\tau_2,\tau_3)\\&= \left\|(\mathbf{Y}_h)_{(3)} - \mathbf{X}_{(3)}\mathbf{M}\right\|_F^2 + \left\|(\mathbf{Y}_m)_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2\\&\quad+\mu\sum_{t=1}^{3}\left\|\mathcal{U}_t - \mathcal{X} + \frac{\zeta_t}{2\mu}\right\|_F^2 + \mu\sum_{t=1}^{3}\left\|\mathcal{V}_t - \nabla_t(\mathcal{U}_t) + \frac{\tau_t}{2\mu}\right\|_F^2\\&\quad+\lambda\sum_{t=1}^{3}\alpha_t\|\mathcal{V}_t\|_{\circledast}\end{aligned} \tag{12}$$

where $\zeta_t, \tau_t (t = 1, 2, 3)$ are the Lagrangian multipliers and μ is the penalty parameter. The following subproblems can be solved iteratively in order to minimize the augmented Lagrangian function (12).

The subproblem with respect to $\mathcal{X}$

$$\begin{aligned}\mathcal{X} = \underset{\mathcal{X}}{\operatorname{argmin}}\ &\left\|(\mathbf{Y}_h)_{(3)} - \mathbf{X}_{(3)}\mathbf{M}\right\|_F^2 + \left\|(\mathbf{Y}_m)_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2\\&+\mu\sum_{t=1}^{3}\left\|\mathcal{U}_t - \mathcal{X} + \frac{\zeta_t}{2\mu}\right\|_F^2.\end{aligned} \tag{13}$$

By setting the derivative of (13) to zero, we can derive the following equation:

$$\mathbf{H}_1\mathbf{X}_{(3)} + \mathbf{X}_{(3)}\mathbf{H}_2 = \mathbf{H}_3 \tag{14}$$

where

$$\begin{aligned}\mathbf{H}_1 &= \mathbf{R}^T\mathbf{R} + 3\mu\mathbf{I}\\\mathbf{H}_2 &= (\mathbf{M})(\mathbf{M})^T\\\mathbf{H}_3 &= \mathbf{R}^T(\mathbf{Y}_m)_{(3)} + (\mathbf{Y}_h)_{(3)}(\mathbf{M})^{\mathrm{T}} + \mu\sum_{t=1}^{3}\left(\mathbf{U}_{t(3)} + \frac{\zeta_{t(3)}}{2\mu}\right)\end{aligned} \tag{15}$$

where $\mathbf{I} \in \mathbb{R}^{S\times S}$ is an identity matrix. The Sylvester equation (14) can be efficiently solved as demonstrated in [26].

The subproblem with respect to $\mathcal{V}_t$

$$\mathcal{V}_t = \underset{\mathcal{V}_t}{\operatorname{argmin}}\,\mu\left\|\mathcal{V}_t - \nabla_t(\mathcal{U}_t) + \frac{\tau_t}{2\mu}\right\|_F^2 + \lambda\alpha_t\|\mathcal{V}_t\|_{\circledast}. \tag{16}$$

We can derive the solution by setting the derivative of Equation (16) to zero

$$\mathcal{V}_t = \text{t-SVT}_{\lambda\alpha_t/2\mu}\left(\nabla_t(\mathcal{U}_t) - \frac{\tau_t}{2\mu}\right) \tag{17}$$

where t-SVT$(\cdot)$ is the tensor singular value thresholding defined in [35].

The subproblem with respect to $\mathcal{U}_t$

$$\mathcal{U}_t = \underset{\mathcal{U}_t}{\operatorname{argmin}}\,\mu\left\|\mathcal{U}_t - \mathcal{X} + \frac{\zeta_t}{2\mu}\right\|_F^2 + \mu\left\|\mathcal{V}_t - \nabla_t(\mathcal{U}_t) + \frac{\tau_t}{2\mu}\right\|_F^2. \tag{18}$$

By setting the derivative of (18) to zero, we can derive the following equation:

$$\begin{aligned}&\left(\mu + \mu\nabla_t{}^{\mathrm{T}}\nabla_t\right)\mathbf{U}_{(t)} =\\&\qquad\mu\mathbf{X}_{(t)} - \frac{\boldsymbol{\zeta}_{(t)}}{2} + \mu\nabla_t{}^{\mathrm{T}}\left(\mathbf{V}_{(t)}\right) + \nabla_t{}^{\mathrm{T}}\left(\frac{\boldsymbol{\tau}_{(t)}}{2}\right)\end{aligned} \tag{19}$$

where $\nabla_t{}^{\mathrm{T}}(\cdot)$ is the transpose operator of $\nabla_t(\cdot)$. By performing Fourier transform on (19), combining convolution theory, the solutions with respect to $\mathcal{U}$ can be calculated as follows:

$$\mathbf{H} = \mathcal{F}(\mathbf{D}_t)^* \odot \mathcal{F}\left(\text{fold}\left(\mu\mathbf{V}_{(t)} + \boldsymbol{\tau}_{(t)}/2\right)\right) \tag{20}$$

$$\mathbf{U}_{(t)} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}\left(\text{fold}\left(\mu\mathbf{X}_{(t)} - \boldsymbol{\zeta}_{(t)}/2\right)\right) + \mathbf{H}}{\mu\mathbf{1} + \mu\left|\mathcal{F}(\mathbf{D}_t)\right|^2}\right) \tag{21}$$

where $\mathcal{F}(\cdot)$ represents the Fourier transform, $\mathcal{F}(\mathbf{D}_t)$ represents the operator $\nabla_t{}^{\mathrm{T}}\nabla_i$ diagonalized by the 3-D fast Fourier transform (FFT) matrix, $\odot$ represents the elementwise multiplication, and $|\cdot|^2$ is the elementwise square operation.

Finally, we update the Lagrange multipliers

$$\zeta_t = \zeta_t + 2\mu(\mathcal{U}_t - \mathcal{X}) \tag{22}$$

$$\tau_t = \tau_t + 2\mu(\mathcal{V}_t - \mathcal{U}_t). \tag{23}$$

Algorithm 1 WLRST for MSI and HSI Fusion

Input: $\mathcal{Y}_h, \mathcal{Y}_m, \mathbf{M}, \mathbf{R}$
set: $k = 0$, choose $\lambda, \mu, \varepsilon > 0, \mathcal{U}_t, \mathcal{V}_t, \zeta_t, \tau_t (t = 1, 2, 3)$
Update iteration:
Update $\mathcal{X}^{(k+1)}$ by solving Equation (14)
$\mathbf{U}_{(t)}^{(k)} \leftarrow \mathbf{X}_{(t)}^{(k+1)}, t = 1, 2, 3$
Update α_t by computing Equation (10)
$\mathcal{V}_{(t)}^{(k+1)} \leftarrow \text{t-SVT}_{\lambda\alpha_t/2\mu}\left(\nabla_t(\mathbf{U}_{(t)}^{(k)}) - \frac{\tau_t}{2\mu}\right), t = 1, 2, 3$
Update $\mathcal{U}^{(k+1)}$ by computing Equation (21)
$\zeta^{(k+1)} \leftarrow \zeta + 2\mu(\mathcal{U}^{(k+1)} - \mathcal{X}^{(k+1)})$
$\tau^{(k+1)} \leftarrow \tau + 2\mu(\mathcal{V}^{(k+1)} - \mathcal{U}^{(k+1)})$
Update iteration: $k \leftarrow k + 1$
until some stopping criterion is satisfied.
Output: $\mathcal{U}_k$

The proposed WLRST fusion method is summarized in Algorithm 1.

The convergence of the ADMM solver applied in Algorithm 1 is nontrivial to analyze due to the nonconvexity introduced by the weighted low-rank regularization. Nevertheless, ADMM can converge for certain classes of nonconvex problems, such as Lipschitz differentiable functions, in which the function is differentiable and its gradient is Lipschitz continuous [40]. In our formulation, the weight update rule is explicitly computable, and the regularization term is Lipschitz differentiable with respect to the optimization variables. Fig. 2 shows the relative error of the proposed WLRST fusion algorithm as a function of iteration times. As shown in Fig. 2, the relative error reaches a stable level close to zero when the number of iterations arrives at 250. Therefore, the ADMM iterations for solving problem (11) can be shown to satisfy the convergence properties. In practice, Algorithm 1 is terminated when the relative change of the primary variable falls below a prescribed threshold (i.e., $1e-5$) or the maximum number of iterations (300) is reached.

III. Experiments

In this section, we compare the proposed WLRST method with five state-of-the-art hyperspectral and multispectral image fusion methods: HySure [24], CSU [23], CSTF [28], LTTR

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[30], WNNSTV [27], and ISGM-Fus [41]. HySure, CSU, and WNNSTV use matrix decomposition techniques for the fusion of HSI and MSI, applying constraints on the abundance matrix. In contrast, CSTF and LTTR are fusion algorithms based on tensor representations for HR-HSIs. CSTF applies sparse constraints to the core tensor, while LTTR imposes a low-rank prior directly on the 4-D tensors of HR-HSIs. ISGM-Fus applies a split-band low-rank prior, which divides the spectrum into correlated subbands and enforces low-rankness within each subband to model spectral correlations. To ensure a fair comparison, we tuned the parameters of all algorithms and evaluated their performance in resolution enhancement from both quantitative and visual perspectives.

A. Experimental Datasets

We consider two scenarios with different land cover categories as simulated datasets, each with distinct geographic and sensor characteristics. We utilize HSIs from Pavia University [42], LaoYu River [43], and Harvard [44] to generate pairs of colocated hyperspectral/MSIs. Specific descriptions of each dataset are given in the following.

1) *The Pavia University dataset:* It was captured using the reflective optics spectrographic imaging system (ROSIS) over the Pavia University, Italy, in 2001. It comprises 610 × 340 pixels and featuring 103 spectral bands spanning a wavelength range of 430–860 nm [42]. The spatial resolution of the acquired HSI is 1.3 m. The scene includes nine types of land cover classes: asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows. Bands with low signal-to-noise ratio (SNR) are excluded, resulting in the utilization of only 93 bands for subsequent analysis. The LR-HSI is simulated by applying a Gaussian filter with a size of 7 × 7 and a standard deviation of 2, followed by downsampling the image by a factor of 4. The HR-MSI is obtained using an IKONOS-like spectral response [25].
2) *The LaoYu River dataset:* It was collected by the OHS satellite over Dianchi Lake and its adjacent LaoYu River wetland parks. The original dataset consists of 491 × 550 pixels with a spatial resolution of 10 m. For our experiments, we selected a subset image of size 480 × 300 pixels from the top-right corner. The wavelength range of the dataset spans from 0.4 to 1 μm. In this region, numerous Metasequoias have been planted along Dianchi Lake, playing a crucial role in purifying the lake's water quality and protecting the ecological environment. After removing the interfering bands, the dataset retains 32 spectral bands for further analysis and experimentation. We simulate the LR-HSI and HR-MSI in the same way as the previous dataset.
3) *The Harvard dataset:* It consists of 50 indoor and outdoor scenes captured under daylight illumination and 27 indoor scenes captured under artificial and mixed lighting conditions. All images were acquired using a commercial hyperspectral camera (Nuance FX, CRI Inc.) equipped with a liquid-crystal tunable filter, yielding 31 spectral bands at 10-nm intervals covering 420–720 nm. The spatial size of each image is 1040 × 1392 pixels. The LR-HSI is generated by first applying an 8 × 8 antialiasing Gaussian filter with a standard deviation of 2 to the HR-HSI and then downsampling by a factor of 8 along both spatial dimensions. The HR-MSI is synthesized by integrating the spectral bands of the ground-truth HR-HSI with a simulated Nikon D700 spectral response function.[1]

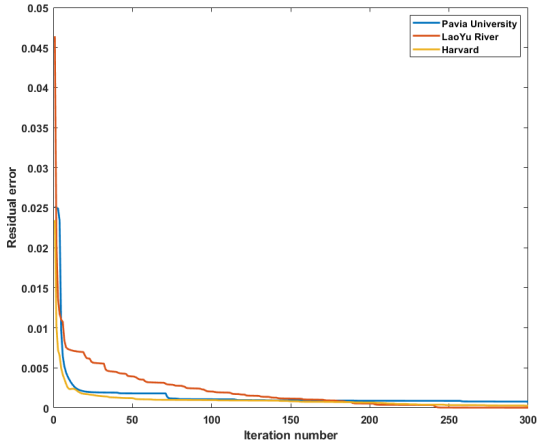


Fig. 2. Relative error of WLRST as a function of iteration number.

B. Image Quality Assessments

The quality of the reconstructed HR HSI is evaluated using six standard metrics: peak signal-to-noise ratio (PSNR) [45], root mean square error (RMSE), spectral angle mapper (SAM) [46], relative dimensionless global error in synthesis (ERGAS), universal image quality index (UIQI) [47], and structural similarity index (SSIM). Higher PSNR indicates greater fidelity to the reference, while lower RMSE denotes more accurate reconstruction. Lower SAM reflects closer spectral similarity, and lower ERGAS indicates better spectral–spatial fidelity. UIQI measures overall image quality via luminance, contrast, and structural correlation, and SSIM assesses perceptual quality by comparing structural information between the reconstruction and the reference. To compute these metrics, all image values were scaled to the range $[0, 255]$.

C. Parameter Analysis

For the proposed WLRST algorithm, we need to set the key parameter λ. The parameter λ represents the weight of the WLRST term and significantly impacts the solution. In the experimental phase, we initially explored a range of values for the parameter. Through a systematic process of trial and error, we adjusted the parameter incrementally to determine the value that minimized the RMSE. Fig. 3 plots the PSNR for the Pavia University, Laoyu River, and Harvard datasets as a function of λ. As depicted in Fig. 3, the PSNR for the Pavia University dataset increases rapidly as λ grows and remains nearly 46 dB

[1] https://www.maxmax.com/spectral_response.htm

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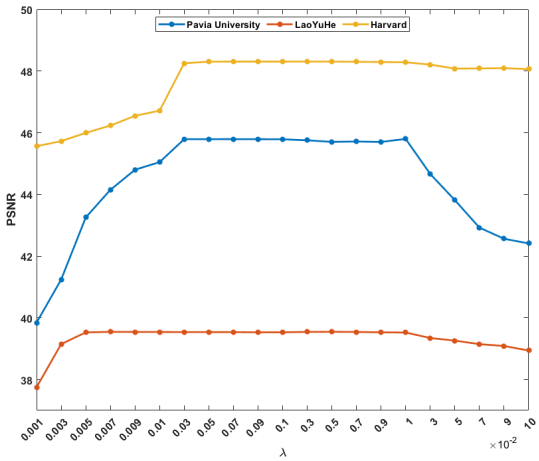


Fig. 3. Performance of the proposed fusion algorithm as a function of λ.

TABLE I

QUANTITATIVE METRICS OF DIFFERENT METHODS EVALUATED ON THE PAVIA UNIVERSITY DATASET

Method	PSNR	RMSE	SAM	ERGAS	UIQI	SSIM
HySure	45.3269	1.5468	1.8893	0.8980	0.9950	0.9925
CSU	33.7098	5.3666	3.2042	3.3393	0.9642	0.9634
CSTF	42.1538	2.0760	2.2193	1.2540	0.9897	0.9816
LTTR	41.7524	2.7440	2.9046	1.6404	0.9866	0.9793
WNNSTV	45.2256	1.5556	**1.6845**	0.9044	0.9950	0.9928
ISGM-Fus	43.0378	2.1235	2.2956	1.2554	0.9913	0.9878
WLRST	**45.7998**	**1.4660**	1.7771	**0.8892**	**0.9955**	**0.9929**

TABLE II

QUANTITATIVE INDEXES OF DIFFERENT METHODS ON THE LAOYUHE DATASET

Method	PSNR	RMSE	SAM	ERGAS	UIQI	SSIM
HySure	37.6068	3.7008	1.5172	0.9115	0.9658	0.9514
CSU	34.4782	5.0613	1.8947	1.1376	0.9539	0.9433
CSTF	38.6512	3.3043	1.3558	0.8038	0.9725	0.9554
LTTR	38.4774	3.5433	1.4225	0.8515	0.9715	0.9511
WNNSTV	37.5205	3.6864	1.5092	0.9068	0.9668	0.9546
ISGM-Fus	37.2512	4.2231	1.6932	0.9383	0.9694	0.9520
WLRST	**39.5481**	**2.9750**	**1.1949**	**0.7208**	**0.9795**	**0.9657**

when λ varies from 3×10^{-4} to 1×10^{-2}, before starting to decline. For the LaoLu River dataset, the PSNR stays between 39 and 40 dB as λ ranges from 5×10^{-5} to 1×10^{-2}, and then, it slightly decreases. For the Harvard dataset, the PSNR increases as λ grows and remains above 48 dB when λ exceeds 3×10^{-4}. Therefore, we set $\lambda = 1 \times 10^{-2}$, $\lambda = 5 \times 10^{-3}$, and $\lambda = 1 \times 10^{-3}$ for the Pavia University, Laoyu River, and Harvard datasets, respectively.

D. Results

Tables I–III present the evaluation metrics for all methods on the Pavia University, LaoYuHe, and Harvard datasets, with the best results highlighted in bold. From Table I, it can be seen that for the Pavia University dataset, the proposed WLRST method achieves the best performance in all evaluation metrics except SAM, where it is the second-best, with results close to the optimal WNNSTV method. For the LaoYuHe and Harvard datasets, Tables II and III show that the proposed WLRST method consistently achieves the best performance in all evaluation metrics. The comparative results across all datasets reveal clear distinctions in performance, which are closely tied to the modeling strategies of each method. HySure, a sparsity-driven HSI–MSI fusion framework, preserves global image structures effectively, as reflected by strong perceptual quality scores, but its limited exploitation of high-order spectral correlations constrains joint spectral–spatial fidelity. CSU, based on spatial–spectral constrained unmixing, benefits from explicit spectral mixing models, yet struggles to capture complex spatial dependencies, leading to weaker reconstruction accuracy in heterogeneous regions. CSTF, which employs coupled spectral–spatial tensor factorization, achieves balanced performance across most metrics, but its fixed-rank constraint limits adaptability to local variations, slightly reducing spectral similarity in challenging scenarios. LTTR leverages a relaxed low TT-rank model to efficiently exploit global low-rank structure; however, constant mode weights hinder its ability to emphasize more informative modes adaptively, affecting the balance between spectral and spatial accuracy. WNNSTV integrates weighted nuclear norm minimization with nonlocal self-similarity and TV regularization, excelling in edge preservation and piecewise smoothness, though its strong spatial regularization can oversmooth subtle spectral variations. In contrast, the proposed WLRST method consistently achieves superior or competitive results across all evaluation metrics. Its adaptive weighting mechanism dynamically adjusts the contribution of each tensor mode, effectively enhancing spectral–spatial fidelity while preserving fine details. Combined with smooth spectral constraints and an efficient nonconvex ADMM-based optimization scheme, WLRST demonstrates both high reconstruction accuracy and robust generalization across datasets with diverse textures and noise characteristics.

TABLE III

QUANTITATIVE INDEXES OF DIFFERENT METHODS ON THE HARVARD DATASET

Method	PSNR	RMSE	SAM	ERGAS	UIQI	SSIM
HySure	46.5883	1.4498	3.8198	1.2342	0.8653	0.9874
CSU	43.3547	2.5352	6.0837	1.2978	0.8617	0.9815
CSTF	45.9342	1.7559	4.6111	1.1554	0.8526	0.9846
LTTR	47.5318	1.2761	3.2506	1.2776	0.8724	0.9903
WNNSTV	44.1046	2.5823	5.2449	1.5957	0.8542	0.9791
ISGM-Fus	47.2706	1.3523	3.2067	1.2086	0.8769	0.9904
WLRST	**48.3050**	**1.2712**	**3.1899**	**0.9414**	**0.8830**	**0.9914**

Figs. 4–6 present visual comparisons of the fused images obtained by different methods on the Pavia University, LaoYuHe, and Harvard datasets, respectively. For the Pavia University dataset, the proposed WLRST method produces false-color composites that are visually closest to the ground truth, with natural color transitions and clear structural boundaries, particularly in high-frequency regions such as building roofs and roads. The single-band and magnified views reveal that WLRST preserves fine spatial details and sharp edges while avoiding ringing artifacts and excessive smoothing. In comparison, WNNSTV and LLTR also maintain relatively good edge sharpness but suffer from slight texture loss and, in the case of WNNSTV, staircase effects. ISGM-Fus preserves the main scene structures reasonably well but still exhibits

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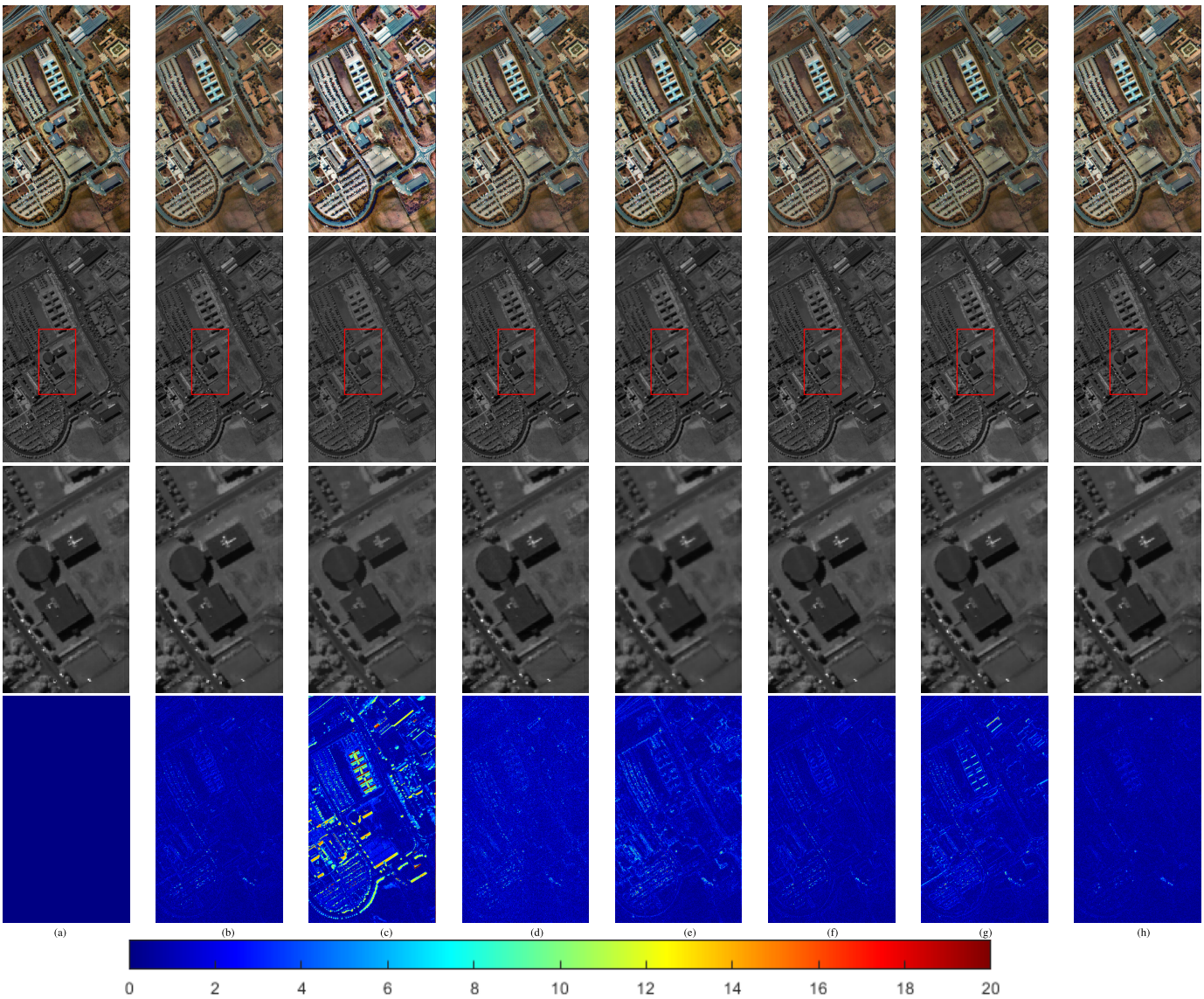


Fig. 4. *First Row:* false-color images of Pavia University obtained by different fusion methods (bands = 60, 25, 4). *Second Row:* reconstructed images at the 68th band. *Third Row:* magnified images of the marked region. *Fourth Row:* corresponding error images at the 20th band. (a) Ground truth. (b) Hysure. (c) CSU. (d) CSTF. (e) LTTR. (f) WNNSTV. (g) ISGM-Fus. (h) WLRST.

slight spectral distortion and detail loss in some textured and edge-rich regions. CSTF and CSU exhibit noticeable striping and checkerboard-like artifacts in detailed regions, while HySure tends to oversmooth, resulting in significant detail loss. This observation is consistent with the error maps, where WLRST shows the lowest intensity (mostly dark blue), while other methods produce stronger errors, particularly along edges and texture areas. For the LaoYuHe dataset, similar trends can be observed. WLRST yields the highest visual fidelity in the false-color composites, accurately recovering color distribution and structural boundaries in complex surfaces. In the single-band images and magnified views, WLRST reproduces sharp and clean edges (e.g., water body boundaries) with smooth gray-level transitions and without blocky or ringing artifacts. WNNSTV and LLTR deliver acceptable reconstructions but suffer from either oversmoothing or slight edge distortion. ISGM-Fus achieves acceptable visual reconstruction, but its results still suffer from slight blurring and limited edge fidelity in regions with complex spatial structures. CSU and CSTF produce blocky and strip-pattern artifacts in complex textures, and HySure again shows excessive blurring. The error maps further confirm the superiority of WLRST, with errors concentrated at a lower level compared to the other methods, demonstrating its strength in both spatial and spectral preservation. For the Harvard dataset, the proposed WLRST method also provides the best visual reconstruction quality among all compared methods. In the false-color composites, WLRST produces results that are the closest to the ground truth, with more natural color reproduction and clearer structural boundaries in both homogeneous and detailed regions. The single-band images and enlarged views further show that WLRST preserves fine spatial structures and edge details while effectively suppressing artifacts and spectral distortion. ISGM-Fus produces visually plausible

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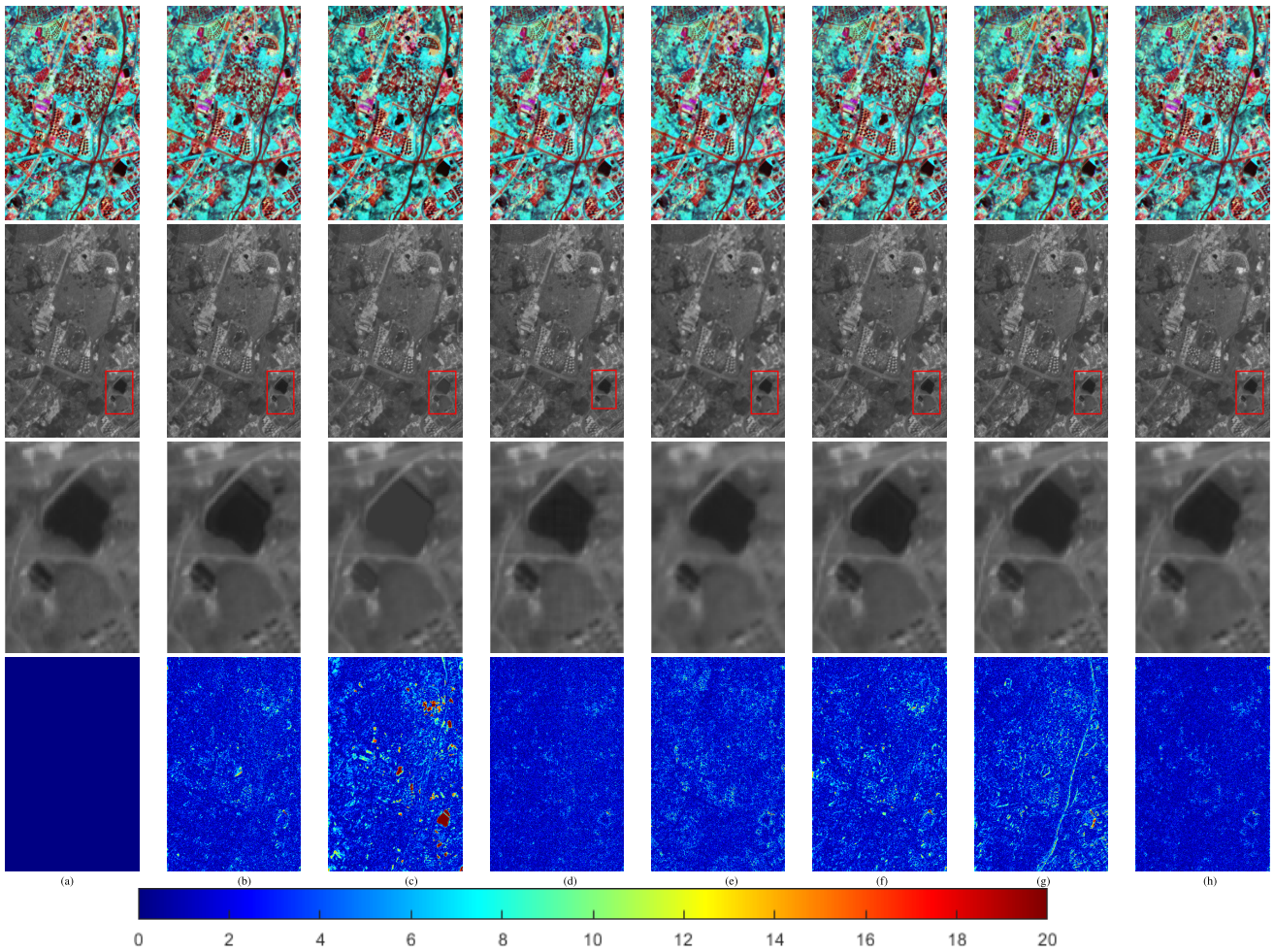


Fig. 5. *First Row:* false-color images of LaoYuHe obtained by different fusion methods (bands = 28, 14, 7). *Second Row:* reconstructed images at the 17th band. *Third Row:* magnified images of the marked region. *Fourth Row:* corresponding error images at the 13th band. (a) Ground truth. (b) Hysure. (c) CSU. (d) CSTF. (e) LTTR. (f) WNNSTV. (g) ISGM-Fus. (h) WLRST.

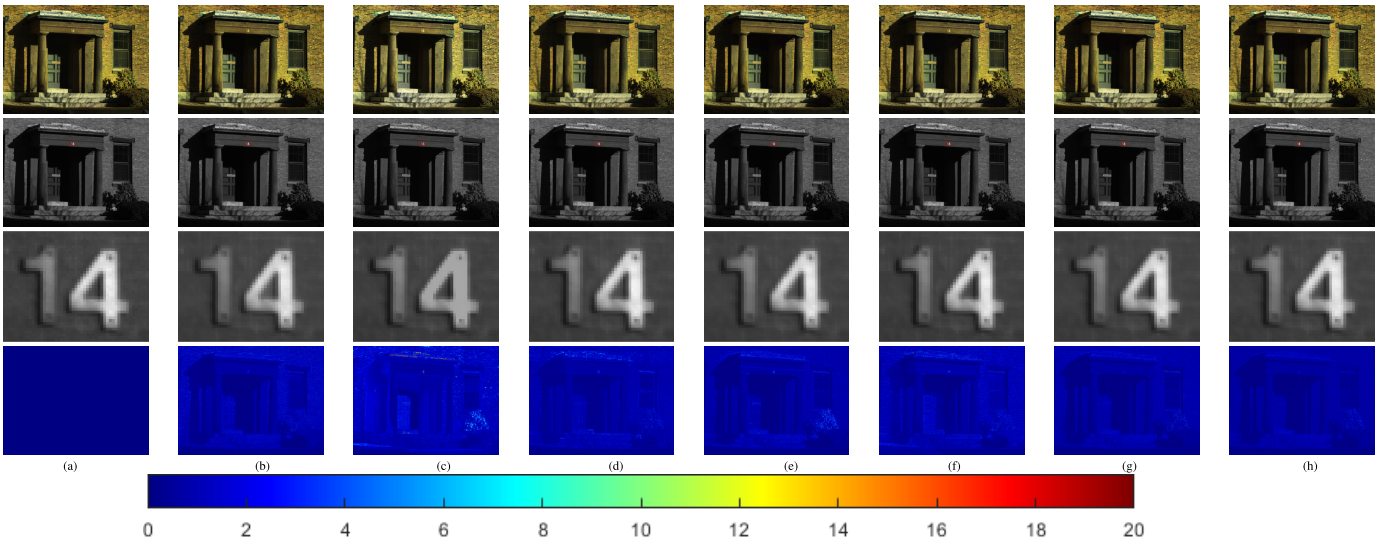


Fig. 6. *First Row:* false-color images of Harvard obtained by different fusion methods (bands = 28, 18, 8). *Second Row:* reconstructed images at the 26th band. *Third Row:* magnified images of the marked region. *Fourth Row:* corresponding error images at the 13th band. (a) Ground truth. (b) Hysure. (c) CSU. (d) CSTF. (e) LTTR. (f) WNNSTV. (g) ISGM-Fus. (h) WLRST.

fused images, but some fine structures and local spectral details are not recovered as accurately as those by WLRST. By contrast, WNNSTV and LLTR achieve relatively competitive visual results but still exhibit slight detail degradation or

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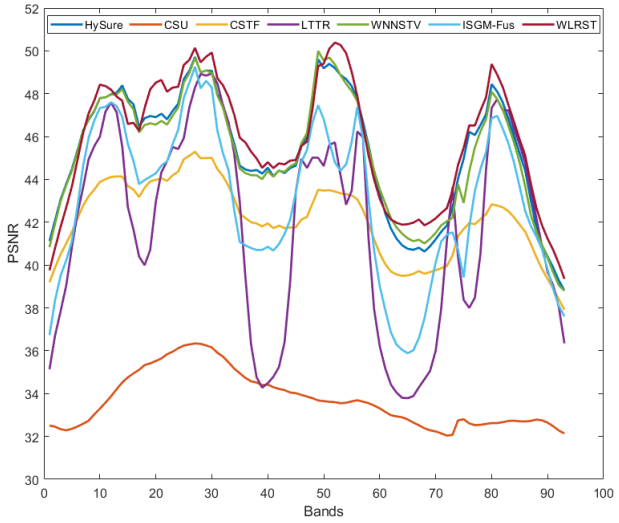


Fig. 7. Bandwise PSNR curves of the fused HR-HSI obtained by different algorithms on the Pavia University dataset.

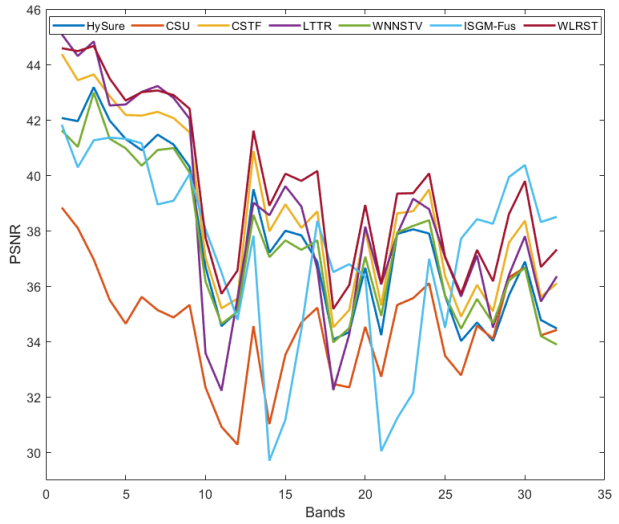


Fig. 8. Bandwise PSNR curves of the fused HR-HSI obtained by different algorithms on the LaoYuHe dataset.

local oversmoothing. CSTF and CSU tend to introduce noticeable artifacts and structural inconsistencies in textured areas, whereas HySure produces overly smooth reconstructions with missing fine details. These observations are further supported by the error maps, where WLRST presents lower error intensity than the other methods, indicating its superior ability to maintain both spatial detail and spectral fidelity. Overall, the qualitative comparisons across two datasets demonstrate that WLRST achieves a favorable balance between detail preservation and noise/artifact suppression, yielding the most visually faithful fusion results with minimal structural and spectral distortion.

Figs. 7–9 present the bandwise PSNR of the fusion results obtained by different algorithms on the three datasets. For the Pavia University dataset, the proposed WLRST method exhibits consistently high PSNR values across almost all spectral bands, with only minor fluctuations, demonstrating its ability to maintain reconstruction quality throughout the

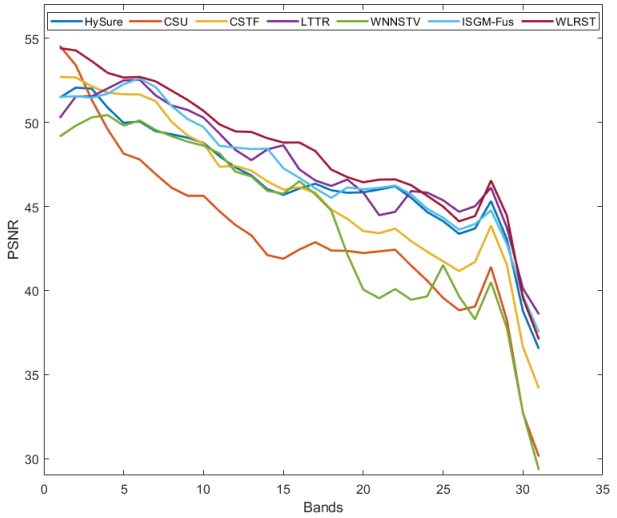


Fig. 9. Bandwise PSNR curves of the fused HR-HSI obtained by different algorithms on the Harvard dataset.

spectrum. In particular, WLRST either achieves the highest or remains very close to the highest PSNR in most bands, indicating stable spectral fidelity and robustness to band-specific variations. ISGM-Fus achieves relatively competitive PSNR values in several bands, but its performance is less stable than that of WLRST. HySure and WNNSTV also show strong performance in several bands, but their curves display larger variance and occasional drops, suggesting less stability in certain spectral regions. LTTR and CSTF, while competitive in some mid-spectrum ranges, suffer from noticeable performance declines in other bands due to their limited adaptability to local spectral–spatial changes. CSU consistently records the lowest PSNR values across the spectrum, indicating limited reconstruction accuracy. For the LaoYuHe dataset, which contains more complex textures and noise, all methods experience larger PSNR fluctuations across bands compared to the Pavia case. Nevertheless, WLRST maintains relatively higher PSNR values in most spectral regions, particularly in the low- and mid-wavelength ranges, reflecting its robustness under more challenging imaging conditions. ISGM-Fus obtains acceptable bandwise PSNR results but shows noticeable fluctuations in some spectral regions. HySure and WNNSTV remain competitive in certain regions but still exhibit sharper drops in performance in specific bands. LTTR and CSTF show moderate results but struggle to sustain high-quality reconstructions in bands where noise or spectral variation is more prominent. Again, CSU remains the lowest performer across almost all bands, highlighting its difficulty in capturing fine spectral–spatial details under complex conditions. For the Harvard dataset, WLRST also achieves the best overall bandwise PSNR performance. Its curve remains consistently high and stable across nearly all spectral bands, indicating reliable spectral fidelity and reconstruction robustness. ISGM-Fus delivers moderate PSNR performance overall but still falls short of WLRST in both accuracy and stability. HySure and WNNSTV are competitive in some bands but show larger fluctuations, while LTTR and CSTF obtain moderate results with less stable performance. CSU yields the lowest PSNR

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TABLE IV
ABLATION STUDY ON THE HARVARD DATASET

Method	PSNR	RMSE	SAM	ERGAS	UIQI	SSIM
TV-only	46.5566	1.4975	3.7019	1.5647	0.8722	0.9884
LR-only	47.7539	1.3811	3.4185	1.0395	0.8785	0.9905
Uniform-weight	47.8436	1.2946	3.2937	1.2620	0.8808	0.9907
WLRST	**48.3050**	**1.2712**	**3.1899**	**0.9414**	**0.8830**	**0.9914**

values in most bands. Overall, the bandwise PSNR curves clearly demonstrate that WLRST not only achieves superior average performance but also offers enhanced stability across the entire spectral range for both datasets. This stability is largely attributable to its adaptive weighting low-rank tensor modeling, which dynamically balances spectral and spatial contributions, mitigating performance degradation in challenging spectral regions.

E. Ablation Experiments

To evaluate the contribution of each component, an ablation study is conducted on the Harvard dataset by comparing the full model with three ablated variants under the same experimental settings. Specifically, TV-only retains only the TV regularization term; LR-only retains only the low-rank term and imposes it on the reconstructed image tensor rather than in the gradient-tensor domain; and Uniform-weight keeps both the low-rank and TV terms but replaces the adaptive modewise weights with equal weights for all three modes. All variants are evaluated using the same parameter tuning protocol and performance metrics to ensure a fair comparison.

As shown in Table IV, the full model (WLRST) consistently achieves the best performance among all variants, indicating that the improvement comes from the joint effect of gradient-domain low-rank modeling, TV regularization, and adaptive mode balancing, rather than from any single component alone. The TV-only variant preserves local smoothness but is limited in capturing global spatial–spectral correlations. The LR-only variant exploits global redundancy and interband correlations, yet is less effective in preserving local edges and fine details. Combining both regularizers yields clear performance gains, confirming that they are complementary. Moreover, the uniform-weight variant performs worse than the full model, showing that equal treatment of all modes cannot fully account for their distinct structural characteristics. By contrast, the proposed adaptive weighting strategy better balances modewise contributions and, thus, improves both spatial detail preservation and spectral fidelity.

IV. CONCLUSION

In this work, we propose a new WLRST model for fusing multispectral and HSIs. By jointly exploring the global low-rank prior and local sparse–smooth constraints in the spatial–spectral tensor domain, the proposed method effectively exploits both interband correlations and spatial structural characteristics. The superior performance of WLRST can be attributed to its integration of weighted low-rank modeling with adaptive sparse–smooth regularization, which effectively balances spatial detail preservation and spectral distortion reduction. Future work will focus on developing more adaptive prior modeling and parameter selection strategies to improve robustness under diverse imaging conditions while exploring more efficient optimization schemes to enhance computational efficiency and scalability for large-scale datasets. The proposed framework will also be extended to more challenging practical scenarios, such as severe noise, spectral variability, and sensor mismatch, as well as to related hyperspectral image restoration tasks, including denoising and inpainting.

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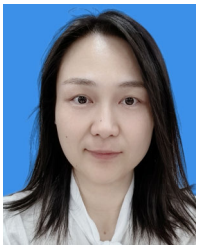
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